Test 1 review day

CMSE 381

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Announcements

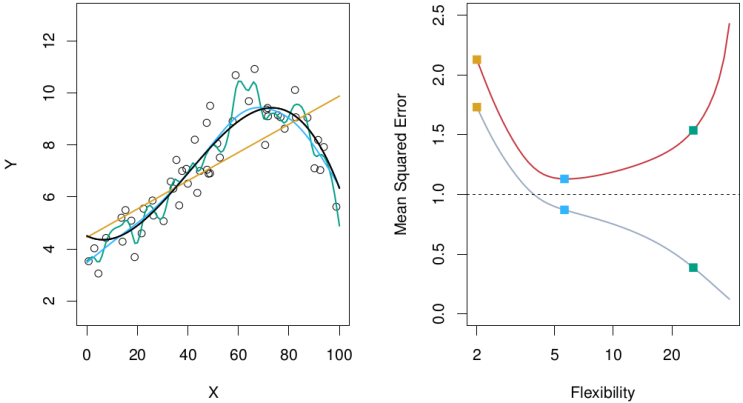
Announcements:

- midterm # 1 on Wednesday
- Project info, honors option posted on course website

Based on the questions submitted

- Bias Variance Trade off
- KNN classification
- How to prepare

Training error vs test error



- Left: Black line is model for simulation of data
- Right side: training in blue/grey, testing in red
- Variance of ε is dashed line
- Point out that training error goes down but test goes up (overfitting)
- Flexibility = degrees of freedom

Bias-variance

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

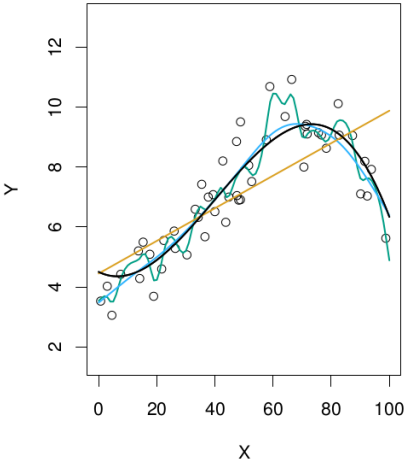
- Proof outside of the scope of class
- $E(y_0 - \hat{f}(x_0))^2$ is expected test MSE at x_0; avg test MSE if repeatedly estimated f with lots of training sets and tested each at x_0
- Computed by averaging $E(y_0 - \hat{f}(x_0))^2$ over all values in the test set

- Eqn says we need an $\hat{f}$ with both low variance and low bias
- Also says error is bounded below by irreduceable error $\mathrm{Var}(\varepsilon)$

Variance

Variance: the amount by which $\hat{f}$ would change if we estimated it using a different training data set.

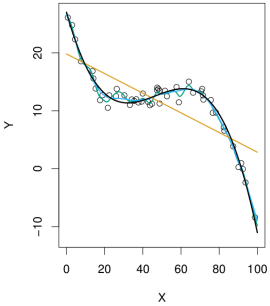
- High variance: small changes in training data result in large changes in $\hat{f}$.
- Example right: green curve more flexible, but also small changes in data set could cause large changes in the computed model.

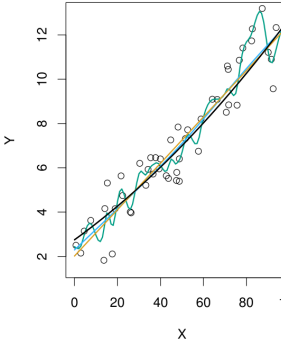


Bias

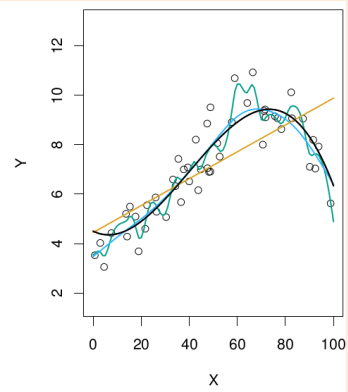
Bias: the error that is introduced by approximating a (complicated) real-life problem by a much simpler model.

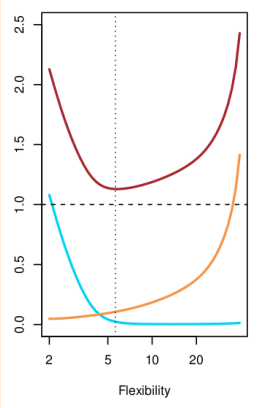
- Example: Linear regression, likely too simple for any real-life problem.
- Figure at left: True f is non-linear, so no good estimate possible with linear regression. (linear regression = high bias)
- Figure at right: True f is linear, so linear regression should do a good job (linear regression = low bias)





Which is which





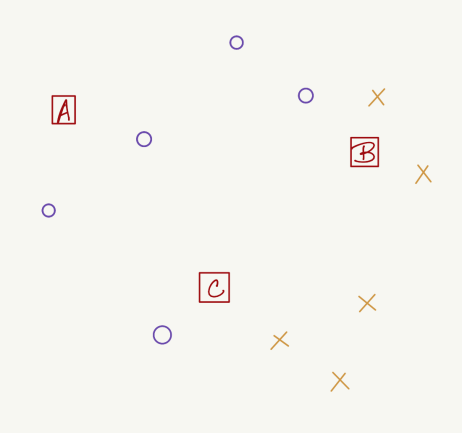
Label the line corresponding to each of the following:

- MSE
- Bias
- Variance of $\hat{f}(x_0)$
- Variance of ε

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Review KNN classifier example

Here label is shown by O vs X. What are the knn predictions for points A, B and C for $k = 1$ or $k = 3$? *Next year, maybe make an example like this with more than two levels, and/or try to decide what locations get which predictions for k=1 in prep for the next slide*



Point	$k = 1$ Prediction	$k = 3$ Prediction
A	*O*	*O*
B	*X*	*X*
C	*O*	*X*

Preparing for exam

- review slides
- write down key concepts equations on the cheatsheet, annotate them
- find pictures, equations, results tables you don't understand. bring to this class or read up in text book.